

MO421 - Introduction to cryptography - Exercise list 2

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1. We say that a polynomial p is positive if for all n , $p(n) > 0$.

A function $f : \mathbb{N} \rightarrow \mathbb{R}_{>0}$ is *negligible* if for every *positive* polynomial $p \in \mathbb{R}[X]$ there is an $n_p \in \mathbb{N}$ such that for all $n > n_p$, it holds that $f(n) < 1/p(n)$.

We denote by $\text{negl}(n)$ the set of negligible functions on n .

For each of the following items, indicate if the statement is true or false. If it is false, present a counterexample (if it is true, you do not need to prove it).

- (a) If $f(n) \in \text{negl}(n)$, then $c \cdot f(n) \in \text{negl}(n)$ for any constant $c \in \mathbb{N}^*$.
 - (b) If $f(n), g(n) \in \text{negl}(n)$, then $f(n) + g(n) \in \text{negl}(n)$.
 - (c) If $f(n), g(n) \in \text{negl}(n)$, then $f(n)/g(n) \in \text{negl}(n)$.
 - (d) If $f(n), g(n) \in \text{negl}(n)$, then $f(n) \cdot g(n) \in \text{negl}(n)$.
 - (e) If $f(n) \in \text{negl}(n)$, then $q(n) \cdot f(n) \in \text{negl}(n)$ for polynomial $q(n)$.
 - (f) Let $f : \mathbb{N} \rightarrow \mathbb{R}_{>0}$ and $g(n) := (f(n))^n$. If $g(n) \in \text{negl}(n)$, then $f(n) \in \text{negl}(n)$.
 - (g) If $f(n) \in \text{negl}(n)$ and $g(n) \notin \text{negl}(n)$, then $f(n) \cdot g(n) \in \text{negl}(n)$.
2. Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function. For the following constructions of a keyed function $G : \{0, 1\}^n \times \{0, 1\}^{n-1} \rightarrow \{0, 1\}^{2n}$ state whether G is a pseudorandom function. If yes, prove it; if not, show a distinguisher.
- (a) $G_k(x) = F_k(0||x)||F_k(0||x)$
 - (b) $G_k(x) = F_k(0||x)||F_k(x||1)$
 - (c) $G_k(x) = F_k(0||x)||F_k(x||0)$
3. [PP10] We are using AES in counter mode for encrypting a hard disk with 1TB of capacity. What is the maximum length of the IV?
4. Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a keyed function. Suppose that there exists an algorithm $\mathcal{A}$ that outputs k when given polynomially many pairs $(x_0, F_k(x_0)), \dots, (x_{\text{poly}(n)}, F_k(x_{\text{poly}(n)}))$. Show that there is a PPT distinguisher D which has non-negligible advantage in distinguishing F from a random function.
5. Consider the following keyed function F . For security parameter n , the key is an $n \times n$ boolean matrix $\mathbf{A}$ and an boolean vector $\vec{b} \in \{0, 1\}^n$. That is, $k = (\mathbf{A}, \vec{b})$. Define $F_k : \{0, 1\}^n \rightarrow \{0, 1\}^n$ as $F_k(\vec{x}) = \mathbf{A} \cdot \vec{x} + \vec{b} \bmod 2$.
- Show an algorithm that can recover k given $n+1$ evaluations of F_k . Then use the previous exercise to show that F is not a pseudorandom function.

6. Let $E : \{0, 1\}^{128} \times \{0, 1\}^{128} \rightarrow \{0, 1\}^{128}$ be a secure block cipher. Consider ECB mode encryption for two-block messages

$$(m_1, m_2) \in (\{0, 1\}^{128})^2$$

defined as

$$\text{Enc}_k(m_1, m_2) = (E_k(m_1), E_k(m_2)).$$

Show that ECB mode is not CPA-secure.

7. AES is used in CTR mode. Encryption of blocks m_i is defined as

$$C_i = m_i \oplus \text{AES}_k(\text{IV} || i).$$

Suppose two different messages are encrypted using the same key and the same IV.

- (a) Show that

$$C_i \oplus C'_i = m_i \oplus m'_i$$

for every block i .

- (b) Explain why this implies that reusing the IV breaks CPA security.

8. Let $F : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function.

Consider the following encryption scheme for two-block messages $(m_1, m_2) \in (\{0, 1\}^n)^2$:

$$\text{Enc}_k(m_1, m_2) = (r, F_k(r) \oplus m_1, F_k(r) \oplus m_2),$$

where $r \leftarrow \{0, 1\}^n$ is chosen uniformly.

Is this scheme CPA-secure? Justify your answer.

References

- [KL21] Jonathan Katz and Yehuda Lindell. *Introduction to Modern Cryptography*. Third edition. CRC Press. 2021.
- [PP10] Christof Paar and Jan Pelzl. *Understanding Cryptography*. First edition. Springer 2010.